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20CS5T01- COMPUTER NETWORKS				

COURSE OUTCOMES :
At the end of the course, student will be able to
CO1: Differentiate network reference models such as OSI, TCP/IP (K2)
CO2: Classify various Data Link Layer protocols such as Error Detection and correction. (K2)
CO3: Distinguish various MAC sub layer Protocols such as ALOHA, CSMA, CSMA/CD (K2)
CO4: Differentiate various Network layer and Transport layer protocols and Its Applications (K2)
CO5: Illustrate various application layer protocols such as WWW and HTTP etc (K2)

UNIT - I :
Data Communication: Components, Data Representation, Data flow (Simplex, Half- duplex and Full-Duplex), Types of connections: Point to Point and Multipoint, Various Categories of Topologies, Categories of Networks, Protocols and Standards, OSI network model, TCP/IP Protocol Suite, Transmission Media (Twisted pair cable, Coaxial cable and Fiber-optic cable).

UNIT - II :
Data Link Layer: Error Detection and Error Correction -Introduction, Block coding: Er- ror Detection, Error Correction, Hamming Distance, Minimum Hamming Distance, Cyclic Codes: Cyclic Redundancy check (CRC), Checksum, Framing, Flow control and Error con- trol.

UNIT - III :
Medium Access Sub Layer: Random Access protocols – ALOHA, Pure ALOHA, Slotted ALOHA, Carrier Sense Multiple Access (CSMA),1-persistent CSMA, Nonpersistent CSMA, p-Persistent CSMA, CSMA/CD, CDMA/CA.
Network Layer: Logical addressing – IPV4 Addresses: Classful and Classless Addressing, Subnetting, Network Address Translation (NAT), IPV6 Addresses-Structure and Address space, Address Mapping: ARP, RARP, BOOTP and DHCP.

UNIT - IV :
Transport Layer: Process to Process Communication, User Datagram Protocol (UDP), UDP Format, uses of UDP, Transmission Control Protocol (TCP), TCP Services, TCP Fea- tures, TCP Segment, Quality of Service, QoS improving techniques: Leaky Bucket and Token Bucket algorithm.

UNIT - V :
Application Layer: Domain Name System (DNS), Domain Name Space, Distribution of Name Space, Remote Logging: TELNET, ELECTRONIC MAIL, SMTP, File Transfer Protocol (FTP), WWW, HTTP.

Text Books:
1. Data Communication and Networking, 5th Edition, Behrouz A. Forouzan, McGrawHill, 2017
2. Computer Networks, 6th Edition, Andrew S. Tanenbaum, Pearson New International Edition, 2021.
3. Data and Computer Communication, 8th Edition, William Stallings, Pearson Prentice Hall India, 2017

Reference Books:
1. Internetworking with TCP/IP, Volume 1, 6th Edition Douglas Comer, Prentice Hall of India. TCP/IP Illustrated, Volume 1, W. Richard Stevens, Addison-Wesley, United States of America.